

Neuromorphic Computing and Memristor-Based Devices: Solid-State Physics and Applications in Intelligent Robotics

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Abstract—The rapid growth of artificial intelligence (AI) and intelligent robotics has intensified the need for computing hardware capable of operating with low latency, low power, and high parallel efficiency. Conventional CMOS architectures remain fundamentally constrained by the von Neumann memory bottleneck, pushing current research toward neuromorphic systems that process information in a manner analogous to biological neural networks. Among the most promising technologies enabling this shift are memristor-based resistive switching devices, whose behavior is rooted in solid-state carrier transport phenomena. In particular, the drift and diffusion of ionic defects, such as oxygen vacancies, govern the formation and rupture of conductive filaments that define each memristive state. By leveraging these physical mechanisms, memristor arrays can store and compute in the same location, enabling fast, event-driven, and energy-efficient operations ideal for robotic perception and adaptive control. This report connects the solid-state physics of drift–diffusion transport with the operation of memristor devices and highlights how these mechanisms support emerging neuromorphic architectures for next-generation intelligent robotic systems.

Index Terms—memristor, neuromorphic computing, carrier drift, carrier diffusion, oxygen vacancies, resistive switching, MIM structure, solid-state electronics, edge AI, intelligent robotics

I. INTRODUCTION

The rapid proliferation of artificial intelligence (AI) and robotics has accelerated the demand for embedded systems capable of real-time responsiveness, ultra-low latency, and low power consumption. Modern intelligent robots rely on perception, decision-making, and actuation pipelines that operate seamlessly, integrating object detection, path planning, adaptive control, and multimodal sensory fusion. These requirements place significant pressure on conventional computing hardware. Traditional CMOS based von Neumann architectures suffer from the well known “memory wall,” where the physical separation between memory and processing units introduces latency, energy loss, and bandwidth limitations. This bottleneck becomes increasingly restrictive as AI workloads scale in complexity. Recent research trends aim to reduce data-movement overhead, enhance computational efficiency, and achieve higher levels of embedded intelligence while simultaneously reducing device footprint [1], [2].

Neuromorphic computing has emerged as a promising alternative to conventional architectures. Inspired by the structure and dynamics of biological neural systems, neuromorphic processors incorporate event-driven spiking neurons, massive parallelism, and co-localized memory and computation.

These features enable energy-efficient and real-time processing, which are essential for robotic applications. Within this paradigm, memristor-based resistive switching devices have gained particular importance as artificial synapses due to their analog programmability, non-volatility, nanoscale dimensions, and compatibility with high-density crossbar arrays.

The operation of memristors is fundamentally rooted in the principles of carrier transport covered in solid-state physical electronics. In particular, carrier drift and diffusion provide the physical basis for how mobile ionic species, such as oxygen vacancies in metal-oxide layers, move under applied electric fields or concentration gradients. These processes directly control the formation and rupture of conductive filaments, which define the low- and high-resistance states of a memristor. By understanding drift- and diffusion-driven transport, we gain insight into the switching dynamics, analog programmability, and non-volatile behavior that make memristor-based devices suitable for neuromorphic computing systems.

The purpose of this report is two-fold. First, it provides a detailed overview of the solid-state physics principles that underlie memristor devices and neuromorphic architectures. Second, it examines their emerging applications in intelligent robotics, including event-driven vision, adaptive motor control, on-device learning, and low-power edge AI processing. The remainder of this paper is organized as follows. Section II introduces the concepts of neuromorphic computing and memristor-based devices. Section III reviews the relevant solid-state physics background and explains how these principles support neuromorphic hardware. Section IV discusses applications in intelligent robotic systems. And Section V concludes the report.

II. NEUROMORPHIC COMPUTING ARCHITECTURE

A. Neuromorphic Computing

1) *Neuromorphic Computing Architecture Explanation:* Neuromorphic computing proposes an alternative hardware architecture inspired directly by the organisation and operation of the biological

brain. In contrast to conventional processors that separate memory and computation, the brain performs both functions simultaneously through networks of neurons and synapses. A biological neural network transmits information through spikes and modifies synaptic connections according to activity, allowing learning, adaptation, and highly efficient parallel processing. This idea motivates the structure of neuromorphic systems, which aim to reproduce these properties at the hardware level

According to IBM Research, the hallmark of a neuromorphic system is the integration of computation and memory alongside event-driven spiking neurons and synapses in a massively parallel architecture [4], [2]. In practice, this means that each “neuron” and “synapse” is implemented in hardware, and spikes (brief electrical pulses) propagate through networks of synapses much as they do in the brain.

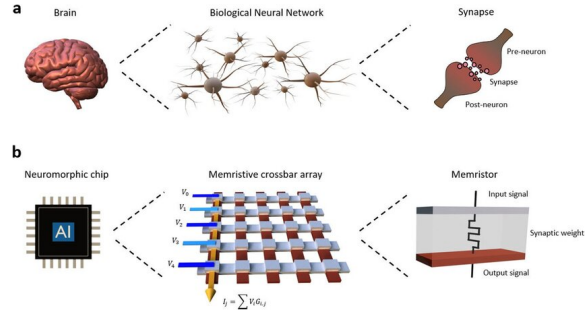


Fig. 1. Neuromorphic Computing Architecture [5]

Figure 1 illustrates the conceptual link between the brain, biological neurons, and synapses, and their hardware equivalents in neuromorphic devices. Each artificial neuron integrates incoming signals and emits output spikes once a threshold is reached, similar to a biological neuron. Synaptic weights, which determine the strength of the connection between neurons, are implemented using devices whose physical state can be gradually tuned. Memristors are particularly promising in this role because their resistance can be modified by electrical pulses, enabling them to behave as analog or multi-level synapses in large arrays [1]. Figure 1(b) depicts this relationship, showing how memristive

crossbar arrays form the basic computational fabric for neuromorphic processors.

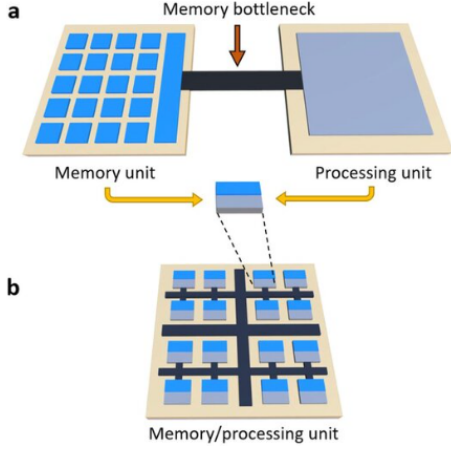


Fig. 2. Neuromorphic Architecture Memory and processing units [5]

A central motivation for neuromorphic architecture is the limitation of the conventional von Neumann model, where memory and processing units are physically separated. As shown in Figure 2(a), data must constantly travel between the memory unit and the processor, creating a memory bottleneck that slows computation and increases energy consumption. This problem becomes more severe as AI workloads grow, since most of the power is spent not on computation but on repeatedly fetching parameters from memory. Neuromorphic systems address this issue by colocating memory and computation within the same physical elements, as illustrated in Figure 2(b). In this architecture, each synaptic device both stores the weight and participates in the computation itself, essentially removing the need for large data transfers.

2) Memristive Explanation:

Memristive crossbar arrays provide a compact and efficient way to implement the synaptic weight matrix of a neural network directly in hardware. As illustrated in Fig. 3(a), the array consists of two orthogonal sets of metal lines, usually referred to as word lines (WLs) and bit lines (BLs), with a vertical stack at each intersection. In the structure shown,

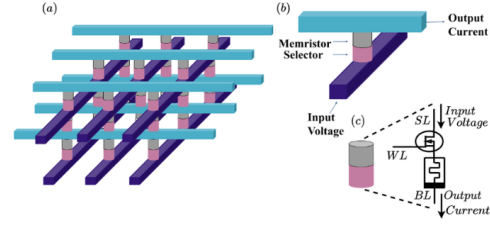


Fig. 3. Memristive crossbar arrays [6]

each cross-point contains a memristor in series with a selector device (often called a 1S1R cell). The selector suppresses unwanted leakage paths, while the memristor encodes the synaptic weight through its programmable conductance [7]. A simplified view of a single cell is shown in Fig. 3(b), and the equivalent circuit representation with source line (SL), WL and BL terminals is depicted in Fig. 3(c).

During a read or inference operation, input voltages V_i are applied to the WLs or SLs, and the resulting output currents I_j are collected at the BLs. Owing to Ohm's law and Kirchhoff's current law, the current at the j -th column is given by

$$I_j = \sum_i G_{ij} V_i, \quad (1)$$

where G_{ij} is the conductance of the memristor located at the intersection of row i and column j . In matrix form, the crossbar naturally performs a matrix-vector multiplication,

$$\mathbf{I} = \mathbf{G} \mathbf{V}, \quad (2)$$

which is the core operation of both artificial neural networks and spiking neuromorphic systems. Because each G_{ij} can be tuned gradually by applying suitable voltage pulses across the corresponding cell, the synaptic weight update required by learning rules such as spike-timing-dependent plasticity (STDP) can be implemented directly at the device level. In this way, the crossbar does not only store the synaptic weights in non-volatile form, but also carries out the weighted-sum computation in a single, massively parallel analog operation [7].

3) *Neuromorphic Computing Architecture Conclusion:* The architectural advantages of neuromorphic computing give to parallelism, event driven operation, and in memory processing, thereby enabling extremely low power and low latency computation. These characteristics are particularly attractive for robotics and edge-AI systems, where fast reaction times, efficient sensory processing, and continuous operation are essential. By avoiding the memory bottleneck and leveraging device level physics for computation, neuromorphic architectures provide a path toward hardware that more closely resembles biological intelligence, both in structure and in efficiency.

4) *Tables Neuromorphic vs von Neumann Architectures:*

TABLE I
COMPARISON BETWEEN NEUROMORPHIC AND VON NEUMANN ARCHITECTURES

Feature	von Neumann	Neuromorphic
Memory vs. compute	Physically separated (memory bottleneck)	Co-located at synapses (in-memory computing)
Data movement	High, continuous memory fetch	Minimal, event-driven communication
Computation model	Sequential, clock-driven	Parallel, spike-based and asynchronous
Power consumption	High, dominated by data transfers	Ultra-low, spikes only on activity
Execution style	Deterministic, instruction-based	Brain-inspired, adaptive behaviour
Scalability	Limited by bandwidth and heat	Scales with number of neurons/synapses
Use in robotics	Requires powerful CPUs/GPUs	Well suited for edge AI and fast reaction

TABLE II
ADVANTAGES AND DISADVANTAGES OF NEUROMORPHIC COMPUTING

Advantages	Disadvantages
Ultra-low power due to event-driven spikes	Device variability and reliability issues in memristive devices
Massively parallel processing of neural networks	Lack of standardised software tools and programming models
In-memory computation (no memory bottleneck)	Training SNNs is more difficult than conventional deep networks
Real-time responsiveness for sensory processing	Limited availability of mature commercial hardware
Biologically inspired learning and adaptation	Integration with CMOS fabrication can be complex
High synapse density using nanoscale memristors	Endurance and retention constraints for long-term operation
Ideal for embedded and robotic edge-AI workloads	Requires new algorithms and learning rules (e.g., STDP)

B. Memristor-Based Devices

Memristors (a contraction of “memory resistor”) constitute a class of two-terminal solid-state devices whose resistance depends on the history of electrical stimuli applied to them. In other words, unlike conventional resistors with fixed values, memristors “remember” past voltage or current and maintain a programmable conductance state even when power is removed. This non-volatility, combined with nanoscale dimensions and analog tunability, makes them uniquely suited to implement synaptic functions in neuromorphic computing.

Memristor technology has emerged as one of the most promising device-level solutions for future neuromorphic systems. Being a two-terminal solid-state device whose resistance is not fixed but depends on the history of the electrical stimulus applied to it. This property, retaining a programmable conductance state without power, makes memristors fundamentally different from conventional resistors or CMOS memory cells. Their behaviour naturally mimics biological synapses, where previous neuronal activity modifies the strength of the connection. By reason, memristors have become central

candidates for hardware implementations of learning and in-memory computing.

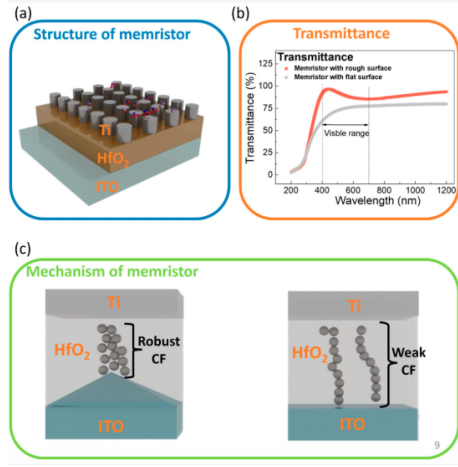


Fig. 4. Memristive crossbar arrays [9]

Figure 4 illustrates a representative structure of an oxide-based memristor. Most devices follow a metal-insulator-metal (MIM) configuration, here shown with a Ti top electrode, a HfO_2 switching layer, and an indium tin oxide (ITO) bottom electrode. The active region is the transition-metal oxide film (HfO_2 in this example), which contains intrinsic crystal defects such as oxygen vacancies. These defects are mobile under an electric field and play a crucial role in defining the device's conductivity. When vacancies accumulate, they form a conductive filament (CF) through the oxide, connecting the two electrodes and lowering the device resistance. When the filament disperses or ruptures, the device returns to a high-resistance state. This reversible process is known as resistive switching.[9]

The transmittance plot in Fig. 4, obtained from recent experimental studies, highlights an additional property of some oxide-based memristors: their optical transparency [8]. Devices fabricated on transparent substrates (e.g., ITO-coated glass) can maintain high transmittance in the visible range even after repeated switching cycles. This allows the development of “invisible” or transparent neuromorphic electronics, which are attractive for wearable and bio-integrated systems.

The mechanism of filament formation is visualized in Fig. 4(c). Under an applied SET voltage, oxygen vacancies drift and aggregate into a dense, robust filament, resulting in a low-resistance state (LRS). Conversely, applying a RESET voltage causes the filament to thin or partially dissolve, leaving only weak clusters of vacancies and restoring the high-resistance state (HRS). Importantly, this filament is not binary; its thickness and continuity can be gradually tuned by adjusting the amplitude, duration, or polarity of electrical pulses. This yields multiple stable conductance levels, enabling analog or multi-bit weight storage, one of the main advantages of memristors over traditional digital memory.

These properties directly support neuromorphic computing. In memristor crossbar arrays, each device acts as an analog synapse whose conductance encodes a synaptic weight. When voltages are applied to the rows of the array, the resulting column currents naturally perform an analog matrix and vector multiplication based on Ohm's and Kirchhoff's laws. Because the memristor both stores the weight and participates in the computation, the architecture avoids the von Neumann memory bottleneck and enables massively parallel, energy-efficient operations. Furthermore, the conductance of each device can be updated locally using voltage pulses, allowing learning rules such as spike-timing-dependent plasticity (STDP) to be realized directly at the device level.

Underlying all these behaviours are the fundamental principles of physical electronics. The operation of a memristor is dictated by the drift and diffusion of ionic species, particularly the motion of oxygen vacancies that accumulate or disperse under an applied electric field. These crystal defects not only define the shape and stability of the conductive filament but also influence the device's switching speed, retention, and variability. At the same time, the changes in energy band structure within the metal-insulator-metal (MIM) stack such as barrier height modulation, trap-assisted transport, and tunnelling pathways govern how electrons move through the device in both the low- and high-resistance states. Variability, endurance, on/off ratio,

and switching energy are not abstract specifications; they emerge directly from these physical processes: charge transport, defect dynamics, and interface physics. In this way, memristors sit precisely at the intersection of materials science, semiconductor device physics, and neuromorphic architecture, forming a clear bridge between the topics studied in solid-state physical electronics and the requirements of emerging AI hardware.

III. SOLID-STATE PHYSICAL ELECTRONICS

A. Carrier Drift and Diffusion

Carrier transport in semiconductors is governed by two fundamental mechanisms: *drift* and *diffusion*, as discussed in [10]. At thermal equilibrium, electrons and holes are never at rest. Instead, they move randomly in all directions due to thermal energy, as typically illustrated by the erratic zig-zag trajectories of carriers undergoing continuous collisions with the lattice and impurities. This thermal motion alone does not produce a net current, because the random velocities statistically cancel each other.

Drift Mechanism. When an external electric field E_x is applied to the semiconductor, the situation changes drastically. Although each individual electron still follows a random path between collisions, the field imposes a small bias on its motion. Electrons experience a force $-qE_x$ in the direction opposite the field, while holes experience a force $+qE_x$ in the same direction as the field. Because the force is continuously interrupted by collisions, electrons do not accelerate indefinitely; instead, they settle into an average velocity known as the *drift velocity*.

The drift velocity is proportional to the applied electric field:

$$v_d = \mu_n E_x, \quad (3)$$

where μ_n is the electron mobility. This mobility reflects how easily electrons can move through the crystal lattice, depending on scattering events with phonons and impurities. Using the momen-

tum-relaxation model, the average velocity in the x -direction becomes

$$\langle v_x \rangle = -\frac{q\tau}{m_n^*} E_x, \quad (4)$$

where τ is the mean free time between collisions and m_n^* is the electron effective mass. This expression shows that even though the instantaneous motion of an electron is chaotic, the average motion over many collisions produces a steady drift opposite to the electric field.

The drift of many electrons per unit volume produces a *drift current density*:

$$J_x = qn\mu_n E_x. \quad (5)$$

When both electrons and holes contribute, their currents add. The total drift current becomes

$$J_x = q(n\mu_n + p\mu_p)E_x = \sigma E_x, \quad (6)$$

where σ is the conductivity of the semiconductor.

Carrier Diffusion. Diffusion is the second fundamental transport mechanism in semiconductors, complementing drift. While drift requires an external electric field, diffusion occurs *even in the absence of any applied field*. It is driven purely by spatial variations in carrier concentration. If one region of a semiconductor contains a high carrier density and the neighbouring region contains fewer carriers, the random thermal motion of electrons and holes causes them to migrate from high to low concentration. This spreading process is analogous to how particles in a gas diffuse to fill a container, and is captured mathematically by Fick's law.

Diffusion Current: For electrons, diffusion produces a current density

$$J_{n,\text{diff}}(x) = qD_n \frac{dn(x)}{dx}, \quad (7)$$

where D_n is the electron diffusion coefficient and $\frac{dn}{dx}$ is the spatial gradient of electron concentration. If the concentration decreases in the $+x$ direction, electrons diffuse toward $+x$, creating a current also directed toward $+x$.

For holes, the current takes the form

$$J_{p,\text{diff}}(x) = -qD_p \frac{dp(x)}{dx}, \quad (8)$$

with the negative sign reflecting the opposite charge and direction conventions.

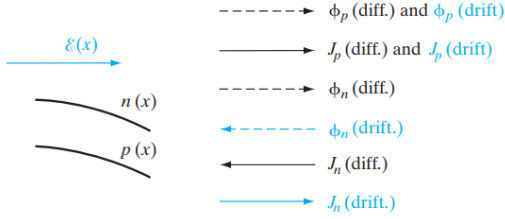


Fig. 5. Drift and diffusion directions for electrons and holes [10]

Figure 5 illustrates how electrons and holes move when both a carrier concentration gradient and an electric field are present. The dashed arrows represent the *particle flow* caused by diffusion: electrons diffuse from regions of high $n(x)$ to low $n(x)$, and holes diffuse from regions of high $p(x)$ to low $p(x)$. The solid arrows represent the resulting *currents*. Because electrons carry negative charge, their diffusion current is opposite to their diffusion flux, while for holes both the flux and current point in the same direction. When an electric field $\mathcal{E}(x)$ is applied, drift superimposes on diffusion: electrons drift opposite the field, and holes drift along the field. Here it clearly shows how diffusion and drift act simultaneously and often in competing directions, with the total current being the sum of both contributions for each carrier type.

Microscopic Interpretation of Diffusion: The physical picture of diffusion is closely tied to the random thermal motion. Even at equilibrium, carriers move in all directions with high thermal velocities. When the carrier concentration is uniform, these random motions cancel statistically, and no net current flows.

However, when a gradient exists the balance is broken:

- more carriers move out of the high-density region than move in,
- fewer carriers move out of the low-density region than move in.

The result is a net particle flux from high concentration to low concentration. This is why the

direction of diffusion current is always opposite to the concentration gradient, consistent with Fick's law.

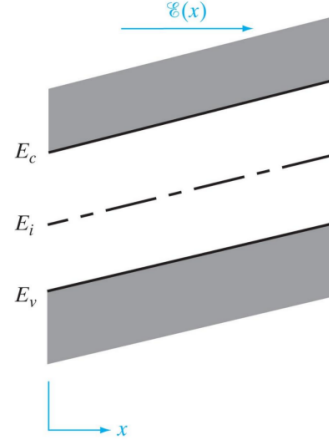


Fig. 6. Energy band diagram of a semiconductor in an electric field [10]

Energy-Band View of Diffusion: Figure 6 shows how an electric field $\mathcal{E}(x)$ tilts the conduction and valence bands. Diffusion, however, can occur with perfectly flat bands (no electric field). When the bands are tilted, drift and diffusion coexist, and the total current becomes:

$$J_n(x) = q\mu_n n(x)\mathcal{E}(x) + qD_n \frac{dn(x)}{dx}, \quad (9)$$

$$J_p(x) = q\mu_p p(x)\mathcal{E}(x) - qD_p \frac{dp(x)}{dx}. \quad (10)$$

This form can be highlighted as one of the most important expressions for semiconductor device physics, since almost all devices involve both fields and concentration gradients at the same time.

Einstein Relation: The diffusion coefficient and mobility are not independent. Random thermal motion links them through the Einstein relation:

$$\frac{D_n}{\mu_n} = \frac{kT}{q} \approx 0.0259 \text{ V at 300K}. \quad (11)$$

This relation is a key result characterizing the connection between thermal energy and transport properties.

Continuity and Time-Dependent Diffusion: Diffusion also appears in the continuity equations, which describe how carrier concentration changes with time. Included the incremental volume analysis:

$$\frac{\partial p(x, t)}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p}, \quad (12)$$

and similarly for electrons. By assuming the current is purely due to diffusion, one obtains the diffusion equation, showing how carrier density smooths out over time:

$$\frac{\partial \delta n}{\partial t} = D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n}, \quad (13)$$

This equation describes transient behaviour such as recombination, carrier spreading, and recovery processes; all key to semiconductor devices in both electronics and optoelectronics.

Summary of Physical Meaning:

- 1) Carriers move randomly due to thermal energy.
- 2) A concentration gradient produces a net flux (diffusion direction arrows).
- 3) Energy bands tilt under an electric field, separating drift from diffusion.
- 4) Drift and diffusion add together to give the total current.
- 5) Time-dependent changes in concentration are governed by the continuity and diffusion equations.

Connection to Memristor-Based Devices. Drift and diffusion form the basis of how memristors change state. In resistive switching oxides such as HfO_2 , the mobile species are not electrons but positively charged oxygen vacancies. Under an applied voltage, these vacancies undergo *drift*, migrating toward one electrode and aggregating into a conductive filament. When a positive set voltage is applied, vacancy drift forms a robust filament, driving the device into a low-resistance state (LRS). Conversely, applying a reset or reverse voltage drives vacancies away or causes them to *diffuse* back into the oxide, rupturing the filament and restoring the high-resistance state (HRS).

Thus, the SET/RESET behaviour of a memristor directly mirrors the drift-diffusion equations covered in solid-state electronics. The spatial redistribution of ionic defects modifies the effective conductivity path between the electrodes, meaning that each memristor state is essentially an equilibrium between drift-dominated filament formation and diffusion-dominated filament dissolution. This physical mechanism enables multilevel conductance tuning, non-volatile memory retention, and the synaptic plasticity required for neuromorphic computing architectures. [11]

IV. APPLICATIONS IN INTELLIGENT ROBOTIC SYSTEMS

Memristor-based neuromorphic hardware is starting to influence several areas of intelligent robotics, mainly because it provides fast, low-power computation directly at the edge, without relying on large GPUs or cloud servers. Modern robots increasingly depend on real-time perception, adaptive control, and on-board learning, and these demands align naturally with the in-memory computing and parallelism offered by memristive synapses.

One of the most promising applications is *event-driven visual processing*. When memristive cross-bars are used to implement spiking neural networks, visual features such as edges, motion, or object location can be extracted directly on-chip with very low latency. This enables robots to react faster to dynamic environments, which is essential for tasks such as obstacle avoidance, target following, or closed-loop visual servoing. Because memristors store the synaptic weights in non-volatile form, the system keeps its learned behaviour even when powered off, unlike traditional accelerators.

Another relevant area is *adaptive motor control*. Neuromorphic controllers built with memristive synapses can update their internal weights using biologically inspired rules such as spike-timing-dependent plasticity (STDP). This makes it possible for robots to tune their control loops on the fly, compensating for actuator wear, payload changes, or uneven terrain. Instead of relying on high-level processors, the learning occurs directly within the device array, reducing computation time

and improving energy efficiency during continuous operation.

A third important application is *energy-efficient on-board inference*. Many robotic platforms; drones, ground robots, humanoids, operate on strict power budgets. Memristive crossbars perform matrix–vector multiplications in a single physical step, enabling classification, sensor fusion, and decision-making at a fraction of the energy required by digital systems. This is especially beneficial in autonomous mobile robots, where every milliwatt saved extends operational time.

V. CONCLUSION

Memristor-based devices and neuromorphic architectures represent a practical and promising path toward the next generation of intelligent robotic systems. By leveraging the principles of physical electronics, particularly drift and diffusion, they provide non-volatile, analog, and highly scalable synaptic elements that conventional CMOS systems cannot match. Their ability to compute and store information in the same physical location eliminates the memory bottleneck that limits real-time robotic perception, decision-making, and control. As robotics continues to advance toward higher autonomy, lower latency, and on-device learning, neuromorphic hardware offers a concrete way to approach biological levels of efficiency. Although challenges remain in device variability, endurance, and large-scale integration, ongoing progress in materials science, device physics, and AI hardware design is steadily closing this gap. Ultimately, memristor-based neuromorphic systems have the potential to become a core technology for future robotic platforms capable of fast, adaptive, and energy-efficient intelligence at the edge.

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